

A proof candidate for the toroidal Elton–Odell theorem

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Abstract

We prove, subject only to standard quoted selection and structural theorems in Banach space theory, that every infinite-dimensional complex normed space contains a sequence of unit vectors which is uniformly separated modulo unimodular scalars. The new ingredient is a rapid flat-block alternative: failure of toroidal separation along an asymptotically monotone basic sequence produces bounded partial sums of successive flat blocks with unimodular external phases. Bounded completeness and strong summability then rule out this bad alternative. This manuscript is written as a proof candidate, with constants and quantifiers exposed for review.

1 Statement and notation

Throughout, scalars are complex. For unit vectors x, y in a normed space, set

$$d_{\mathbb{T}}(x, y) = \inf_{|\theta|=1} \|x - \theta y\|.$$

The target statement is the following.

Theorem 1.1 (Toroidal Elton–Odell). *Let X be an infinite-dimensional complex normed space. Then there are $\varepsilon > 0$ and unit vectors $(x_n)_{n=1}^{\infty} \subset X$ such that*

$$\|x_n - \theta x_m\| \geq 1 + \varepsilon \quad (n \neq m, |\theta| = 1).$$

Equivalently, $d_{\mathbb{T}}(x_n, x_m) \geq 1 + \varepsilon$ for all $n \neq m$.

We shall prove the result first for Banach spaces and then pass from a normed space to its completion.

2 Standard external ingredients

We use the following standard facts.

(S1) James’ non-distortion theorem for c_0 and ℓ_1 : if a Banach space contains an isomorphic copy of c_0 , respectively ℓ_1 , then it contains copies with Banach–Mazur distance arbitrarily close to 1 from c_0 , respectively ℓ_1 .

(S2) Every infinite-dimensional Banach space contains a basic sequence. A basic sequence whose closed span is reflexive is boundedly complete.

(S3) Every basic sequence has an asymptotically monotone block basis [3, Lemma 2.4]. Every seminormalized weakly null sequence has an asymptotically monotone basic subsequence; see Barroso’s strengthening of the Bessaga–Pelczynski selection principle [1, Lemma 3.1].

(S4) Rosenthal’s ℓ_1 theorem: if a Banach space contains no copy of ℓ_1 , then every bounded sequence has a weak-Cauchy subsequence.

(S5) Rosenthal’s c_0 theorem: a Banach space contains no copy of c_0 if and only if every nontrivial weak-Cauchy sequence has a strongly summing subsequence [6, Corollary 1.1]; see also [5].

Here a basic sequence (s_n) is *strongly summing* if it is weak-Cauchy and whenever scalars (c_n) satisfy

$$\sup_N \left\| \sum_{n=1}^N c_n s_n \right\| < \infty,$$

the scalar series $\sum_n c_n$ converges.

3 The easy c_0 and ℓ_1 cases

We record a small transfer estimate.

Lemma 3.1 (Almost isometric transfer). *Let E be a normed space and let $(u_n) \subset S_E$ satisfy*

$$\inf_{n \neq m} d_{\mathbb{T}}(u_n, u_m) \geq \sigma > 1.$$

Suppose that $T : E \rightarrow X$ is linear and, for some $0 < \delta < 1$,

$$(1 - \delta)\|u\| \leq \|Tu\| \leq (1 + \delta)\|u\| \quad (u \in E).$$

Set $x_n = Tu_n / \|Tu_n\|$. Then

$$d_{\mathbb{T}}(x_n, x_m) \geq \frac{(1 - \delta)\sigma - 2\delta}{1 + \delta} \quad (n \neq m).$$

In particular, if $\delta > 0$ is small enough, the right-hand side is strictly larger than 1.

Proof. Fix $n \neq m$ and $|\theta| = 1$. Put $a_n = \|Tu_n\|$ and $a_m = \|Tu_m\|$. Then $1 - \delta \leq a_n, a_m \leq 1 + \delta$, and

$$\|Tu_n - \theta Tu_m\| \geq (1 - \delta)\|u_n - \theta u_m\| \geq (1 - \delta)\sigma.$$

On the other hand,

$$\begin{aligned} \|Tu_n - \theta Tu_m\| &= \|a_n x_n - \theta a_m x_m\| \\ &\leq a_n \|x_n - \theta x_m\| + |a_n - a_m| \\ &\leq (1 + \delta)\|x_n - \theta x_m\| + 2\delta. \end{aligned}$$

Rearranging and then taking the infimum over $\theta \in \mathbb{T}$ proves the estimate. $\square$

Lemma 3.2. *If a Banach space X contains a copy of c_0 or a copy of ℓ_1 , then X satisfies Theorem 1.1.*

Proof. In ℓ_1 , the unit vector basis satisfies

$$\|e_n - \theta e_m\|_1 = 2 \quad (n \neq m, |\theta| = 1).$$

By James' non-distortion theorem and Lemma 3.1, the conclusion transfers to every Banach space containing ℓ_1 .

In c_0 , define

$$u_n = e_1 + \cdots + e_n - e_{n+1} \quad (n \geq 1).$$

Then $\|u_n\|_\infty = 1$. If $n < m$, the coordinates 1 and $n+1$ give

$$\|u_n - \theta u_m\|_\infty \geq \max\{|1 - \theta|, |-1 - \theta|\} = \max\{|1 - \theta|, |1 + \theta|\}.$$

Since

$$\inf_{|\theta|=1} \max\{|1 - \theta|, |1 + \theta|\} = \sqrt{2},$$

we have $d_{\mathbb{T}}(u_n, u_m) \geq \sqrt{2}$ for $n \neq m$. James' non-distortion theorem and Lemma 3.1 transfer this to every Banach space containing c_0 . $\square$

4 Asymptotically monotone bases and flat blocks

Let (e_n) be a normalized basic sequence. For $N \in \mathbb{N}$, define the tail initial-projection constant

$$M_N = \sup \frac{\left\| \sum_{i=p}^q a_i e_i \right\|}{\left\| \sum_{i=p}^r a_i e_i \right\|},$$

where the supremum is taken over all $N \leq p \leq q \leq r$, all finite scalar families $(a_i)_{i=p}^r$, and all choices with nonzero denominator. The sequence (e_n) is called *asymptotically monotone, in the tail-initial sense used here*, if

$$M_N \longrightarrow 1.$$

This is the only projection property used below. It is implied by the usual condition $\|P_n\| \rightarrow 1$ for the ordinary partial-sum projections, because the vectors to which P_q is applied below have zero coordinates before their first support index. Equivalently, for every $\gamma > 0$ there is N such that, whenever $N \leq p \leq q \leq r$,

$$\left\| \sum_{i=p}^q a_i e_i \right\| \leq (1 + \gamma) \left\| \sum_{i=p}^r a_i e_i \right\|.$$

Definition 4.1 (Flat blocks). A *flat block* of (e_n) is a nonzero finite vector

$$d = \sum_{i \in F} \lambda_i e_i, \quad F \subset \mathbb{N} \text{ finite and nonempty}, \quad |\lambda_i| = 1.$$

Gaps in F are allowed. If $0 < \eta < 1$, an η -flat unit block is a unit vector of the form

$$b = \frac{d}{\|d\|},$$

where d is a flat block and

$$|||d|| - 1| \leq \eta.$$

For two finite-support vectors u, v , write $u < v$ if $\max \text{supp } u < \min \text{supp } v$.

5 The rapid flat-block alternative

Proposition 5.1 (Rapid flat-block alternative). *Let (e_n) be a normalized asymptotically monotone basic sequence. Suppose that, for every $\varepsilon > 0$, there is no infinite sequence of unit vectors $(x_n) \subset \overline{\text{span}}\{e_n : n \in \mathbb{N}\}$ satisfying*

$$d_{\mathbb{T}}(x_n, x_m) > 1 + \varepsilon \quad (n \neq m).$$

Then there are successive flat blocks

$$d_1 < d_2 < d_3 < \dots$$

and phases $\alpha_j \in \mathbb{T}$ such that

$$\|d_j\| \longrightarrow 1$$

and

$$\sup_N \left\| \sum_{j=1}^N \alpha_j d_j \right\| < \infty.$$

Proof. Choose positive sequences (η_j) and (γ_j) with

$$0 < \eta_j, \gamma_j < 2^{-j-10}$$

so fast decreasing that, if

$$\rho_j = \sum_{k=j}^{\infty} (2\eta_k + \gamma_k),$$

then

$$\rho_{j+1} \leq \eta_j \quad (j \geq 1).$$

This is possible, for instance, by taking η_j and γ_j super-exponentially decreasing.

We recursively choose integers m_j and finite families B_j of flat unit blocks. Suppose m_j has been chosen so far out that the tail after m_j has initial-projection constant at most $1 + \gamma_j$; that is, whenever $m_j \leq p \leq q \leq r$,

$$\left\| \sum_{i=p}^q a_i e_i \right\| \leq (1 + \gamma_j) \left\| \sum_{i=p}^r a_i e_i \right\|. \quad (1)$$

Such choices are possible because (e_n) is asymptotically monotone.

Consider all η_j -flat unit blocks supported after m_j . Under the standing assumption, this class contains no infinite subfamily whose distinct members are pairwise at $d_{\mathbb{T}}$ -distance greater than $1 + \eta_j$. Hence the greedy construction of such a family terminates, and there is a finite maximal family

$$B_j = \{b_1^{(j)}, \dots, b_{r_j}^{(j)}\}$$

of η_j -flat unit blocks supported after m_j such that

$$d_{\mathbb{T}}(b_i^{(j)}, b_k^{(j)}) > 1 + \eta_j \quad (i \neq k),$$

and maximal with respect to this property. For each $i \leq r_j$, choose a representing flat block $d_i^{(j)}$ such that

$$b_i^{(j)} = \frac{d_i^{(j)}}{\|d_i^{(j)}\|}, \quad |||d_i^{(j)}|| - 1| \leq \eta_j. \quad (2)$$

Because B_j is finite, we may then choose m_{j+1} larger than every support appearing in B_j and also far enough that the analogue of (1) holds with $j+1$ in place of j .

Maximality gives the following covering fact: if u is any η_j -flat unit block supported after m_j , then there are $i \leq r_j$ and $\theta \in \mathbb{T}$ such that

$$|||b_i^{(j)} - \theta u|| \leq 1 + \eta_j. \quad (3)$$

Indeed, if no such pair (i, θ) existed, then u could be added to B_j while preserving pairwise $d_{\mathbb{T}}$ -separation greater than $1 + \eta_j$.

Fix $L \in \mathbb{N}$. We construct flat blocks

$$z_L^{(L)}, z_{L-1}^{(L)}, \dots, z_1^{(L)}$$

backwards. Choose any $i_L \leq r_L$ and set

$$z_L^{(L)} = d_{i_L}^{(L)}.$$

By (2),

$$|||z_L^{(L)}|| - 1| \leq \eta_L \leq \rho_L. \quad (4)$$

Suppose $j < L$ and $z_{j+1}^{(L)}$ has already been constructed. Inductively it is a flat block supported after every member of B_j and satisfies

$$|||z_{j+1}^{(L)}|| - 1| \leq \rho_{j+1}. \quad (5)$$

Since $\rho_{j+1} \leq \eta_j$, the vector

$$u_{j+1} = \frac{z_{j+1}^{(L)}}{\|z_{j+1}^{(L)}\|}$$

is an η_j -flat unit block supported after m_j . By (3), choose $i_j \leq r_j$ and $\theta_j \in \mathbb{T}$ such that

$$|||b_{i_j}^{(j)} - \theta_j u_{j+1}|| \leq 1 + \eta_j. \quad (6)$$

Define

$$z_j^{(L)} = d_{i_j}^{(j)} - \theta_j z_{j+1}^{(L)}. \quad (7)$$

Because the two summands have successive disjoint supports and unimodular coefficients on their supports, $z_j^{(L)}$ is again a flat block.

We now estimate its norm. Put

$$s = \|d_{i_j}^{(j)}\|, \quad t = \|z_{j+1}^{(L)}\|.$$

By (2) and (5),

$$|s - 1| \leq \eta_j, \quad |t - 1| \leq \rho_{j+1}.$$

150 Using (6),

$$\begin{aligned}
151 \quad \|z_j^{(L)}\| &= \|sb_{i_j}^{(j)} - \theta_j t u_{j+1}\| \\
152 \quad &\leq \|b_{i_j}^{(j)} - \theta_j u_{j+1}\| + |s-1| + |t-1| \\
153 \quad &\leq 1 + 2\eta_j + \rho_{j+1}.
\end{aligned} \tag{8}$$

154 For the lower estimate, $d_{i_j}^{(j)}$ is the initial interval projection of $z_j^{(L)}$ onto the interval hull of its
155 first block. Since the whole support lies after m_j , (1) gives

$$156 \quad \|d_{i_j}^{(j)}\| \leq (1 + \gamma_j) \|z_j^{(L)}\|.$$

157 Therefore

$$158 \quad \|z_j^{(L)}\| \geq \frac{1 - \eta_j}{1 + \gamma_j} \geq 1 - \eta_j - \gamma_j. \tag{9}$$

159 Combining (8) and (9), and recalling the definition of ρ_j , we obtain

$$160 \quad |||z_j^{(L)}|| - 1| \leq 2\eta_j + \gamma_j + \rho_{j+1} = \rho_j.$$

161 This closes the backward induction.

162 Unwinding (7), for each L there are indices $i_j^{(L)} \leq r_j$ and phases $\alpha_j^{(L)} \in \mathbb{T}$ such that

$$163 \quad z_1^{(L)} = \sum_{j=1}^L \alpha_j^{(L)} d_{i_j^{(L)}}^{(j)}. \tag{10}$$

164 The estimate just proved gives

$$165 \quad \|z_1^{(L)}\| \leq 1 + \rho_1 \quad (L \in \mathbb{N}). \tag{11}$$

166 If $N \leq L$, then

$$167 \quad \sum_{j=1}^N \alpha_j^{(L)} d_{i_j^{(L)}}^{(j)}$$

168 is the initial interval projection of $z_1^{(L)}$ onto the first N selected block levels. Applying (1) with
169 $j = 1$ gives

$$170 \quad \left\| \sum_{j=1}^N \alpha_j^{(L)} d_{i_j^{(L)}}^{(j)} \right\| \leq (1 + \gamma_1)(1 + \rho_1) =: C. \tag{12}$$

171 For each fixed j , the index $i_j^{(L)}$ takes values in the finite set $\{1, \dots, r_j\}$, while $\alpha_j^{(L)}$ lies in the
172 compact circle $\mathbb{T}$. A diagonal compactness argument gives a subsequence $L_k \rightarrow \infty$ such that, for
173 every fixed j ,

$$174 \quad i_j^{(L_k)} = i_j \quad \text{eventually,} \quad \alpha_j^{(L_k)} \rightarrow \alpha_j \in \mathbb{T}.$$

175 Set

$$176 \quad d_j = d_{i_j}^{(j)}.$$

177 Then $d_1 < d_2 < \dots$, and (2) gives

$$178 \quad |||d_j|| - 1| \leq \eta_j,$$

179 so $\|d_j\| \rightarrow 1$. Fixing N in (12) and passing to the limit along L_k yields

$$180 \quad \left\| \sum_{j=1}^N \alpha_j d_j \right\| \leq C.$$

181 Since N is arbitrary, the desired boundedness of partial sums follows. $\square$

182 6 Excluding the bad alternative

183 **Lemma 6.1** (Boundedly complete case). *Let X contain a boundedly complete basic sequence.*
 184 *Then X satisfies Theorem 1.1.*

185 *Proof.* Assume toward a contradiction that no sequence of unit vectors in X is toroidally $(1 + \varepsilon)$ -
 186 separated for any $\varepsilon > 0$. Pass to a normalized asymptotically monotone block basis (e_n) of a
 187 boundedly complete basic sequence. The block basis remains boundedly complete.

188 Apply Proposition 5.1 to (e_n) . We obtain successive flat blocks d_j with $\|d_j\| \rightarrow 1$ and phases
 189 $\alpha_j \in \mathbb{T}$ such that

$$190 \quad \sup_N \left\| \sum_{j=1}^N \alpha_j d_j \right\| < \infty.$$

191 Since (d_j) is a seminormalized block basic sequence of a boundedly complete basis, (d_j) is bound-
 192 edly complete. Hence the series $\sum_j \alpha_j d_j$ converges. Its terms must then tend to 0, contradicting

$$193 \quad \|\alpha_j d_j\| = \|d_j\| \rightarrow 1.$$

194 Thus the desired toroidally separated sequence must exist. $\square$

195 **Lemma 6.2** (Strongly summing case). *Let (s_n) be a nontrivial strongly summing basic sequence*
 196 *in a Banach space X . Then X satisfies Theorem 1.1.*

197 *Proof.* Assume, again toward a contradiction, that no sequence of unit vectors in X is toroidally
 198 $(1 + \varepsilon)$ -separated for any $\varepsilon > 0$.

199 Because (s_n) is weak-Cauchy and nontrivial, it is not norm Cauchy. Hence there is $r > 0$ and
 200 successive pairs

$$201 \quad p_1 < q_1 < p_2 < q_2 < \dots$$

202 such that

$$203 \quad \|s_{p_n} - s_{q_n}\| \geq r \quad (n \geq 1). \quad (13)$$

204 Define

$$205 \quad y_n = \frac{s_{p_n} - s_{q_n}}{\|s_{p_n} - s_{q_n}\|}.$$

206 Then (y_n) is normalized and weakly null: for every $f \in X^*$, the scalar sequence $f(s_n)$ is Cauchy,
 207 so $f(s_{p_n}) - f(s_{q_n}) \rightarrow 0$, and the denominators in the definition of y_n are bounded below by r .
 208 By the asymptotically monotone selection principle, pass to an asymptotically monotone basic
 209 subsequence, still denoted (y_n) . Relabel the associated pairs as (p_n, q_n) .

210 Apply Proposition 5.1 to (y_n) . We obtain successive flat blocks

$$211 \quad d_j = \sum_{n \in F_j} \lambda_n y_n, \quad |\lambda_n| = 1, \quad F_1 < F_2 < \dots,$$

212 and phases $\alpha_j \in \mathbb{T}$ such that

$$213 \quad \sup_N \left\| \sum_{j=1}^N \alpha_j d_j \right\| < \infty. \quad (14)$$

214 Define scalars (a_n) by

$$215 \quad a_n = \alpha_j \lambda_n \quad (n \in F_j), \quad a_n = 0 \quad (n \notin \bigcup_j F_j).$$

216 Thus every nonzero a_n has modulus 1, and there are infinitely many nonzero a_n .

217 Let K be the basis constant of (y_n) . From (14), the partial sums at block endpoints are
 218 bounded. Every ordinary partial sum $\sum_{n=1}^m a_n y_n$ is an initial coordinate projection, with respect
 219 to (y_n) , of one of those bounded block-end partial sums. Therefore

$$220 \quad B := \sup_m \left\| \sum_{n=1}^m a_n y_n \right\| < \infty. \quad (15)$$

221 Now expand in the original strongly summing sequence. For each n with $a_n \neq 0$,

$$222 \quad a_n y_n = \frac{a_n}{\|s_{p_n} - s_{q_n}\|} s_{p_n} - \frac{a_n}{\|s_{p_n} - s_{q_n}\|} s_{q_n}.$$

223 Define scalar coefficients (c_k) by

$$224 \quad c_{p_n} = \frac{a_n}{\|s_{p_n} - s_{q_n}\|}, \quad c_{q_n} = -\frac{a_n}{\|s_{p_n} - s_{q_n}\|},$$

225 and $c_k = 0$ at all other indices, with the convention that both displayed coefficients are 0 if $a_n = 0$.

226 We claim that the vector partial sums $\sum_{k=1}^m c_k s_k$ are bounded. If $q_n \leq m < p_{n+1}$, the partial
 227 sum equals $\sum_{i=1}^n a_i y_i$, hence has norm at most B . If $p_n \leq m < q_n$, the partial sum differs from
 228 $\sum_{i=1}^{n-1} a_i y_i$ by

$$229 \quad \frac{a_n}{\|s_{p_n} - s_{q_n}\|} s_{p_n},$$

230 whose norm is at most $\sup_i \|s_i\|/r$ by (13). Since a weak-Cauchy sequence is bounded, this proves

$$231 \quad \sup_m \left\| \sum_{k=1}^m c_k s_k \right\| < \infty. \quad (16)$$

232 Strong summability of (s_n) implies that the scalar series $\sum_k c_k$ converges. But this is impos-
 233 sible. Let $M = \sup_i \|s_i\| < \infty$. For every n with $a_n \neq 0$,

$$234 \quad |c_{p_n}| = \frac{1}{\|s_{p_n} - s_{q_n}\|} \geq \frac{1}{2M}.$$

235 There are infinitely many such n , so the terms of the scalar series $\sum_k c_k$ do not tend to zero. This
 236 contradicts convergence. Therefore the assumed failure of toroidal separation is impossible. $\square$

7 Assembly in Banach spaces

Proposition 7.1. *Every infinite-dimensional complex Banach space satisfies Theorem 1.1.*

Proof. Let X be infinite-dimensional. If X contains c_0 or ℓ_1 , Lemma 3.2 applies. Hence assume that X contains neither c_0 nor ℓ_1 .

If X is reflexive, then by the basic sequence theorem X contains a basic sequence. Its closed span is reflexive, so the basis is boundedly complete. Lemma 6.1 applies.

It remains to consider the case where X is nonreflexive and contains neither c_0 nor ℓ_1 . Choose a bounded sequence in X with no weakly convergent subsequence. Since X contains no ℓ_1 , Rosenthal's ℓ_1 theorem gives a weak-Cauchy subsequence. This weak-Cauchy subsequence is nontrivial, because otherwise the original bounded sequence would have a weakly convergent subsequence. Since X contains no c_0 , Rosenthal's c_0 theorem gives a strongly summing subsequence. Lemma 6.2 applies.

Thus every case gives the desired toroidally separated sequence. $\square$

8 Passage from Banach spaces to normed spaces

Proof of Theorem 1.1. Let X be an infinite-dimensional complex normed space and let $\widehat{X}$ be its completion. By Proposition 7.1, there are $\delta > 0$ and unit vectors $(y_n) \subset \widehat{X}$ such that

$$d_{\mathbb{T}}(y_n, y_m) \geq 1 + \delta \quad (n \neq m).$$

Choose $0 < a < \delta/8$. Since X is dense in $\widehat{X}$, choose $u_n \in X$ with

$$\|u_n - y_n\| < a \quad (n \geq 1).$$

Set $x_n = u_n/\|u_n\|$. If $a < 1/2$, then $\|x_n - y_n\| \leq 2a$ for all n . Hence, for $n \neq m$ and $|\theta| = 1$,

$$\|x_n - \theta x_m\| \geq \|y_n - \theta y_m\| - \|x_n - y_n\| - \|x_m - y_m\| \geq 1 + \delta - 4a > 1 + \delta/2.$$

Thus $(x_n) \subset S_X$ is toroidally $(1 + \delta/2)$ -separated. This proves the theorem for normed spaces. $\square$

9 Reviewer checklist

The proof above reduces the theorem to the rapid flat-block alternative and then eliminates its bad alternative in the two structural cases supplied by bounded completeness and strong summability. The most important places for external review are:

(R1) The greedy maximal-family step in Proposition 5.1: finite maximality follows from the absence of an infinite pairwise $> 1 + \eta_j$ family.

(R2) The backward recursion in Proposition 5.1: the rapid choice $\rho_{j+1} \leq \eta_j$ is what keeps $z_{j+1}/\|z_{j+1}\|$ inside the relevant flat-block class.

(R3) The lower estimate (9): it uses only initial interval projections on sufficiently far tails; gaps inside flat blocks are harmless because missing coefficients are zero.

(R4) The transition from block-end partial sums to all partial sums in Lemma 6.2: this uses the basis constant of the selected difference sequence.

(R5) The scalar contradiction in Lemma 6.2: infinitely many coefficients c_{p_n} have modulus bounded below by $(2 \sup_i \|s_i\|)^{-1}$.

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